11. Credit Score Classification

Aim

To develop a machine learning model for Credit Score Classification and classify customers into different credit score categories.

Software Used

Google Colab

Procedure

1. Import the required Python libraries.
2. Create/load the credit score dataset.
3. Preprocess the data.
4. Separate features and target variable.
5. Split the dataset into training and testing sets.
6. Train a Random Forest Classifier.
7. Predict the credit score category.
8. Calculate accuracy.
9. Display the confusion matrix.

Code

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# Sample Credit Score dataset

X = np.array([

    [25, 30000, 2, 650],

    [35, 50000, 5, 720],

    [45, 70000, 8, 780],

    [23, 25000, 1, 580],

    [30, 45000, 4, 680],

    [50, 90000, 10, 820],

    [28, 35000, 3, 620],

    [40, 60000, 7, 750],

    [55, 100000, 12, 850],

    [26, 28000, 2, 600],

    [38, 55000, 6, 710],

    [48, 80000, 9, 800]

])
```

```
# 0 = Poor, 1 = Average, 2 = Good
y = np.array([
    1, 2, 2, 0, 1, 2,
    0, 2, 2, 0, 1, 2
])
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
# Create model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
# Train
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Credit Score Classification")
print("Accuracy:", accuracy)
# Confusion Matrix
```

```
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

ConfusionMatrixDisplay(

    confusion_matrix=cm,

    display_labels=["Poor", "Average", "Good"]

).plot()

plt.title("Credit Score Classification")

plt.show()
```

OUTPUT

Credit Score Classification

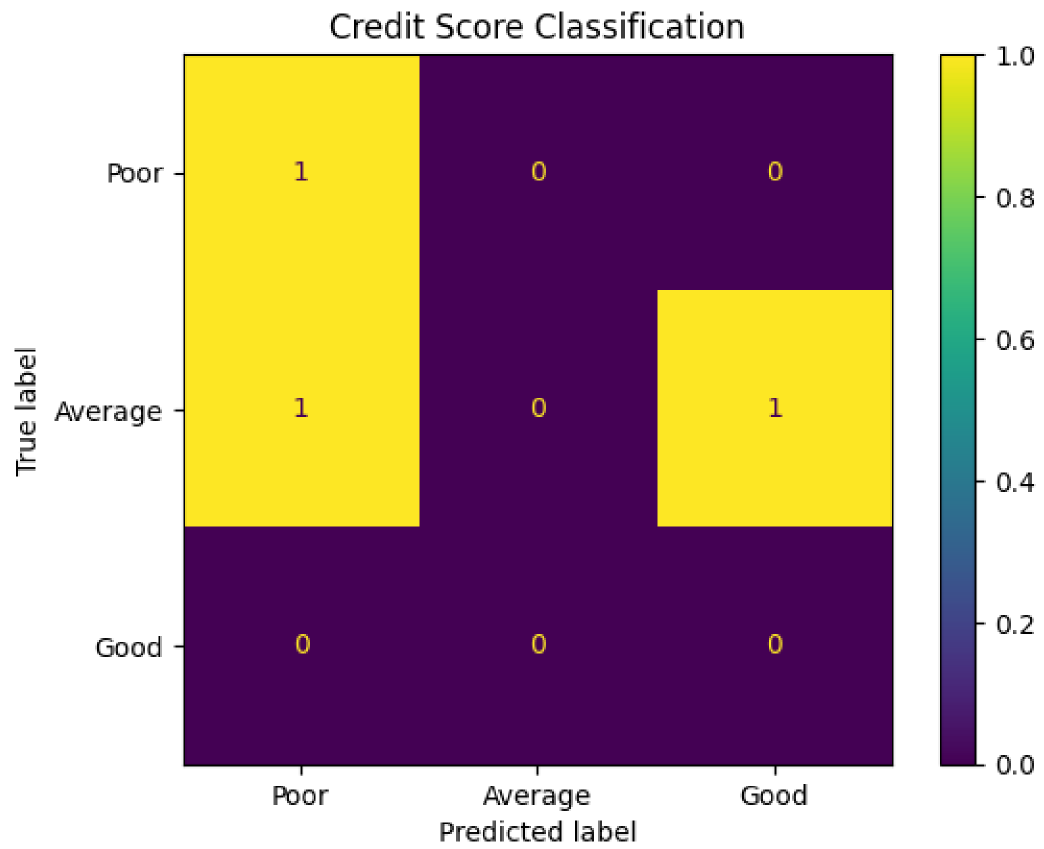
Accuracy: 0.3333333333333333

Confusion Matrix:

[[1 0 0]

 [1 0 1]

 [0 0 0]]



Result

Thus, the Credit Score Classification model was successfully implemented using Random Forest in Python.